

## FULL-LENGTH ORIGINAL RESEARCH

# Extent of neocortical resection and surgical outcome of epilepsy: Intracranial EEG analysis

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### SUMMARY

**Purpose:** Intracranial electroencephalography (EEG) monitoring is an important process in the presurgical evaluation for epilepsy surgery. The objective of this study was to identify the ideal resection margin in neocortical epilepsy guided by subdural electrodes. For this purpose, we investigated the relationship between the extent of resection guided by subdural electrodes and the outcome of epilepsy surgery.

**Methods:** Intracranial EEG studies were analyzed in 177 consecutive patients who had undergone resective epilepsy surgery. We reviewed various intracranial EEG findings and resection extent. We analyzed the relationships between the surgical outcomes and intracranial EEG factors: the frequency, morphology, and distribution of ictal-onset discharges, the propagation speed, and the time lag between clinical and intracranial ictal onset. We also investigated whether the extent of resection, including

the area showing ictal rhythm and various interictal abnormalities—such as frequent interictal spikes, pathologic delta waves, and paroxysmal fast activity—influenced the surgical outcome.

**Results:** Seventy-five patients (42%) were seizure free. A seizure-free outcome was significantly associated with a resection that included the area showing ictal spreading rhythm during the first 3 s or included all the electrodes showing pathologic delta waves or frequent interictal spikes. However, subgroup analysis revealed that the extent of resection did not affect the surgical outcome in lateral temporal lobe epilepsy.

**Conclusions:** The extent of resection is closely associated with surgical outcome, especially in extratemporal lobe epilepsy. Resection that includes the area with total pathologic delta waves and frequent interictal spikes predicts a good surgical outcome.

**KEY WORDS:** Intracranial EEG, Epilepsy, Surgical outcome, Extent of resection, Interictal abnormalities.

Among patients with medically refractory epilepsy, perhaps half are candidates for focal resective surgery (Thadani & Taylor, 2007). Many studies of the prognostic factors for surgical outcomes have been published during the last decade. Major determinants of outcome are complete resection of magnetic resonance imaging (MRI)–visible lesions or the epileptogenic zone (Tassi et al., 2002; Cohen-Gadol et al., 2004; Fauser et al., 2004), underlying pathology (Berkovic et al., 1995; Tonini et al., 2004), the presence of a visible lesion on MRI (Hennessy et al., 2001; Tonini et al., 2004; Yun et al., 2006), and location of seizure focus (Télez-Zenteno et al., 2005; Thadani & Taylor, 2007). Although temporal lobe resection for mesial temporal lobe epilepsy (mTLE) is the most frequently performed surgical pro-

cedure and usually predicts a favorable prognosis, surgical outcome in cases of neocortical epilepsy is generally less satisfactory than for patients with mTLE (Engel et al., 2003; Tonini et al., 2004; Télez-Zenteno et al., 2005; Spencer & Huh, 2008). In terms of evaluation techniques, a concordance between diagnostic tests such as electroencephalography (EEG), MRI, ictal single-photon emission computed tomography (SPECT), or positron emission tomography (PET) is associated with a good surgical outcome (Thadani et al., 2004; Tonini et al., 2004; Yun et al., 2006).

Intracranial EEG is one of the most important procedures in planning surgery and achieving a good surgical outcome in resective epilepsy surgery. The objectives of intracranial EEG are to define interictal abnormalities and the ictal-onset zone, and to map the cortical function (Sperling, 2003). On the basis of these results, physicians can determine the extent of the surgical resection. The surgical outcome depends strongly on the identification and complete resection of a well-defined epileptogenic zone. Therefore, the extent of the resection may contribute to the

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surgical outcome (Widdess-Walsh et al., 2007). However, there are no consistent guidelines for resection based on intracranial EEG in neocortical epilepsy.

In mTLE, seizures are generated within a network of temporal lobe structures (e.g., limbic networks) or extratemporal structures (e.g., insular–frontal–opercular regions) (Henry et al., 1993; Bertram et al., 1998; Bartolomei et al., 2001; Bernasconi et al., 2004; Mueller et al., 2004; Lin et al., 2007). However, the nature of the epileptogenic zone network is unclear in neocortical epilepsy. An understanding of the neocortical epileptogenic network should allow the complete resection of the neocortical epileptogenic zone. Interpreting intracranial EEG is one of the processes that can extend our understanding of the neocortical epileptogenic network.

In this retrospective study, we attempted to identify, using surgical prognostic factors, the intracranial EEG findings that help to define the epileptogenic zone in focal neocortical epilepsy. We investigated the relationship between the extent of neocortical resection—based on the electrographic characteristics of intracranial EEG—and surgical outcomes.

## METHODS

### Patients

We retrospectively reviewed the records of consecutive patients who had undergone resective epilepsy surgery and intracranial EEG monitoring with subdural electrodes for intractable localization-related epilepsy between 1995 and 2003. All the patients had been followed for more than 2 years and had undergone a presurgical evaluation that included MRI, postoperative pathologic evaluation, scalp and intracranial video-EEG monitoring, PET, and interictal and ictal SPECT, if available. Intracranial electrodes were arranged in grids and strips in various combinations. We included only patients who had neocortical ictal onsets, and, therefore, excluded patients with radiologic evidence of hippocampal sclerosis or mesial temporal ictal onsets by intracranial EEG monitoring.

### Intracranial EEG analysis

Intracranial EEG recordings of the patients were made with a 128-channel Telefactor Beehive Horizon digital video monitoring system. The locations of the grids and strips were determined based on clinical, imaging, and scalp EEG data obtained during a noninvasive evaluation. In patients with a lesion demonstrated on MRI, subdural electrodes were placed over the lesion and the surrounding area, including the eloquent cortex. In patients in whom MRI revealed no lesion, the locations of the subdural arrays were guided by the results of ictal scalp EEG, PET, and ictal SPECT studies and semiology. In patients with TLE, intracranial electrodes always included multiple strips reaching to the parahippocampal gyrus to sample mesial temporal

activity to determine the site from which seizure onset originated. A more widespread coverage of the neocortex was made in these patients without demonstrable lesions.

We analyzed the various intracranial EEG features of each patient. The data were analyzed with both bipolar and referential montages. Ictal onset was defined as a paroxysmal sustained rhythmic EEG change differentiated from the background EEG and interictal waves. We also included diverse ictal-onset patterns such as low-amplitude fast and attenuation/suppression pattern in addition to the rhythmic waves and spikes. The ictal-onset zone distribution and propagation time were classified into several groups to compare diverse ictal-onset patterns, and we defined the classification criteria to represent the different patterns most effectively. We classified the ictal-onset zone by location (temporal or extratemporal) and distribution (focal, regional, or widespread). A focal onset was defined as one involving fewer than five adjacent electrode contacts, and regional onset as involving five or more adjacent contacts. Widespread ictal onset was defined as involving more than 20 adjacent electrodes. Ictal onset was sometimes observed in multifocal or noncontiguous electrodes, and we classified this pattern as widespread distribution regardless of the number of involved electrodes. The ictal-onset discharges were classified as delta to theta, alpha, and beta to gamma ranges according to their frequencies and were divided into sinusoidal waves and spikes or sharp waves according to their morphology. Propagation speed was classified as “rapid” when the propagation time from the ictal-onset electrodes to the adjacent electrodes was <1 sec and “slow” when >1 sec. We also investigated the time lag (seconds) between clinical and electrographic ictal onset.

### Surgery and extent of resection

Patients underwent the appropriate surgery performed by one surgeon about 1 week after intracranial EEG monitoring. Only two patients could not undergo the resective surgery: one patient whose ictal-onset area corresponds exactly to the Broca's area could not undergo surgery, and the other patient, who had bihemispheric multifocal ictal-onset areas underwent corpus callosotomy. The extent of resection was determined by considering the localization of ictal onset, interictal spikes, and functional brain areas (motor, sensory, and language cortices) elicited by electrical stimulation. We tried to resect all of the MRI-visible lesion or area of ictal onset, persistent pathologic delta slowing, >1 Hz frequent spikes, and intermittent gamma wave by intracranial EEG, but we could not perform complete resection in some patients because these areas are often multifocal and widespread, and often included portion of eloquent area. During the surgical procedure in patients with TLE, we tried to spare hippocampus for better function outcome, but nine patients whose seizures showed rapid spread to hippocampus or were simultaneously recorded from the hippocampus

and neocortical area underwent hippocampal resection as a part of a tailored temporal lobectomy procedure.

We investigated the relationship between the extent of the resection, including the area showing ictal onset and spreading ictal rhythm, and the surgical outcome. We grossly divided the patients into three groups according to the extent of resection based on ictal onset and spreading ictal rhythm: group I included patients who underwent partial resection of the electrodes in which ictal discharge occurred during the first 3 sec after ictal onset; group II included patients who underwent resection that included all the electrodes in which ictal onset occurred during the first 3 sec; and group III included patients who underwent resection that included all electrodes in which ictal onset occurred during the first 5 sec. We also analyzed whether resection that included various interictal abnormalities (delta waves, frequent interictal spikes, and paroxysmal fast activity) influenced the surgical outcome (Fig. 1). We included interictal spikes above a frequency of  $\frac{1}{5}$  Hz (more than one spike per 5 sec) and divided them into the following three categories: interictal spikes of  $\frac{1}{5}$  to  $\frac{1}{3}$  Hz, of  $\frac{1}{3}$  to 1 Hz, or of  $>1$  Hz. In patients with spikes with two or more kinds of frequency, we classified them in the category appropriate to the faster frequency. We defined pathologic delta waves as persistent arrhythmic or rhythmic slow waves in the delta range and paroxysmal fast activity as paroxysmal EEG activities in the gamma range ( $>35$  Hz).

### Outcomes and statistical analysis

Surgical outcomes were classified according to the method of Engel and colleagues (Engel et al., 1993). We compared the extent of resection, each of the EEG features, and the MRI findings of the seizure-free group (class I group) and the non-seizure-free group (class II–IV group) by univariate and multivariate analyses. The outcomes for each variable were analyzed with a chi-square test, Student's *t*-test, and logistic regression, using the SPSS statistical package (SPSS, Chicago, IL, U.S.A.), version 12.0. *p*-Values of  $<0.05$  were considered to indicate a statistically significant difference.

## RESULTS

### Patients

One hundred seventy-seven patients were included in the study (121 men and 56 women). The median age at epilepsy onset was 12.5 years (range 0.25–36 years). The median age at surgery was 27.9 years (range 11–51 years). The median latency between epilepsy onset and epilepsy surgery was 15.4 years (range 1–50 years). Sixty-six (37.3%) patients had neocortical TLE and 111 patients had extra-TLE (61 frontal, 23 parietal, and 27 occipital), which was classified according to the ictal-onset zone. Sixty-six patients (37.3%) had a visible lesion on MRI. The average follow-up period was 8.24 years (range 2–15 years). There was no significant difference in the follow-up period

between the class I (seizure-free) group (8.09 years) and the class II–IV group (8.33 years). Seventy-five patients (42.4%) achieved a seizure-free outcome. In the nonseizure-free group, 42 achieved a class II outcome and 35 achieved a class III outcome. Twenty-five patients achieved a class IV outcome.

### Intracranial EEG features

In the distribution of ictal onset, 12 patients (6.8%) had wide ictal onset and the others (165 patients, 93.2%) had regional or focal ictal onset. The beta or gamma range was the most common ictal-onset frequency (101 patients, 57.1%), and of the different morphologies, sinusoidal waves (117 patients, 66.1%) were more common than spikes or sharp waves (60 patients, 33.9%). Pathologic delta waves were present in 97 patients (54.8%), frequent interictal spikes ( $>\frac{1}{5}$  Hz) were present in 89 patients (50.3%), and paroxysmal fast activity was present in 20 patients (11.3%). Twenty-two patients (12.5%) had rapid ictal spreading patterns.

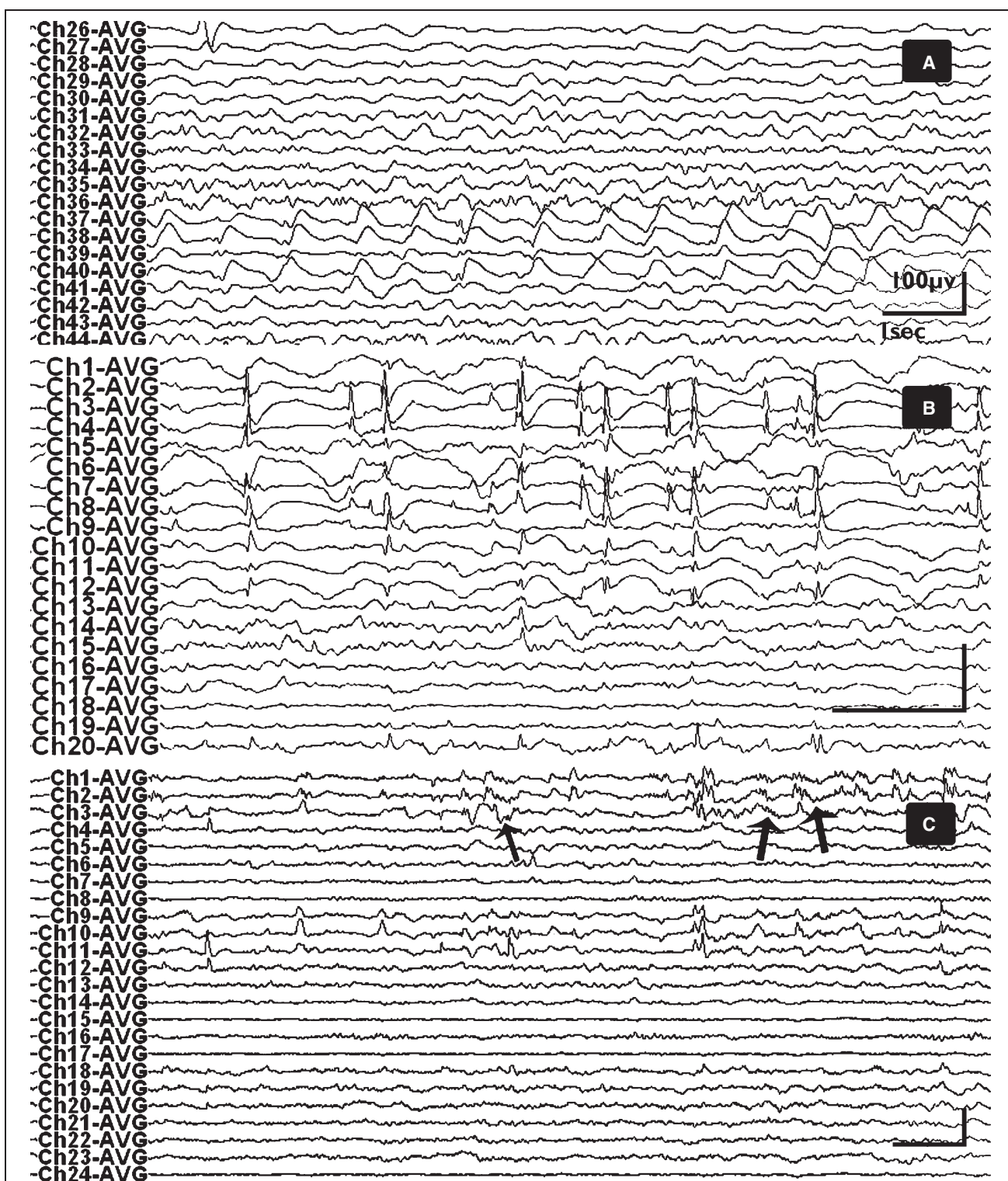
### Surgical outcomes and various predictors

We used statistical analysis to identify the predictors of postoperative seizure-free outcome (Table 1). Based on the results of invasive evaluations, slow propagation ( $p = 0.014$ ) and focal or regional ictal onset ( $p = 0.008$ ) were associated with seizure-free outcomes. However, neither the morphology of the ictal-onset discharges nor the location of ictal onset influenced the surgical outcome. A fast frequency of ictal-onset discharges predicted a tendency to a better surgical outcome (seizure-free rate: 34.1% in delta to theta, 31.3% in alpha, and 49.5% in beta to gamma), but this was not statistically significant ( $p = 0.084$ ). The mean time lag between clinical and electrographic ictal onset was 14.21 sec (range 0–92 sec). There was no significant difference in the mean time lag between the seizure-free and non-seizure-free groups (mean time lag: 12.31 sec in the seizure-free group and 15.62 sec in the non-seizure-free group,  $p = 0.151$ ).

One hundred sixty-nine patients (95.5%) had pathologic abnormalities. Among these patients, histopathologic analyses showed cortical dysplasia in 103 (60.9%), cerebral contusion or stroke in 16 (9.47%), tumor in 10 (5.9%), gliosis in 13 (7.7%), and other abnormalities in 27 (15.9%). There was no pathologic difference between patients with neocortical TLE and those with extra-TLE, and postoperative pathology did not influence the surgical outcome. Conversely, the presence of a visible MRI lesion was significantly associated with a good surgical outcome ( $p < 0.001$ ).

### Surgical outcome and extent of resection

The seizure-free outcome correlated with the extent of resection. Group I consisted of 73 patients, group II consisted of 21 patients, and group III consisted of 83 patients. There was a significant difference in the seizure-free



**Figure 1.**

Examples of interictal electroencephalography (EEG) abnormalities. (A) Persistent rhythmic delta waves are seen in channels 37–40. Frequent interictal spikes (B) and paroxysmal fast activity in channels 1–3 (arrow, C).

*Epilepsia* © ILAE



**Table 1. Surgical outcome and various predictors**

|                                                                                                                                  | No. | Seizure-free (%) | p-Value   |
|----------------------------------------------------------------------------------------------------------------------------------|-----|------------------|-----------|
| EEG                                                                                                                              |     |                  |           |
| Location of ictal onset                                                                                                          |     |                  |           |
| Temporal                                                                                                                         | 66  | 28 (42.4)        | 0.991     |
| Extratemporal                                                                                                                    | 111 | 47 (42.3)        |           |
| Distribution of ictal onset                                                                                                      |     |                  |           |
| Focal/Regional                                                                                                                   | 165 | 75 (45.4)        | 0.008*    |
| Widespread                                                                                                                       | 12  | 0 (0)            |           |
| Propagation speed                                                                                                                |     |                  |           |
| Slow                                                                                                                             | 155 | 71 (45.8)        | 0.014*    |
| Rapid                                                                                                                            | 22  | 4 (18.2)         |           |
| Ictal-onset morphology                                                                                                           |     |                  |           |
| Sinusoidal waves                                                                                                                 | 117 | 54 (46.2)        | 0.155     |
| Spike or sharp waves                                                                                                             | 60  | 21 (35)          |           |
| Ictal-onset frequency                                                                                                            |     |                  |           |
| Beta to gamma                                                                                                                    | 101 | 50 (49.5)        | 0.084     |
| Alpha                                                                                                                            | 32  | 10 (31.3)        |           |
| Delta to theta                                                                                                                   | 44  | 15 (34.1)        |           |
| MRI                                                                                                                              |     |                  |           |
| Visible lesion                                                                                                                   | 66  | 41 (62.1)        | 0 < .001* |
| No lesion                                                                                                                        | 111 | 34 (30.6)        |           |
| Pathology                                                                                                                        |     |                  |           |
| Cortical dysplasia                                                                                                               | 103 | 43 (41.3)        | 0.386     |
| Contusion/Stroke                                                                                                                 | 16  | 6 (37.5)         |           |
| Tumor                                                                                                                            | 10  | 6 (60.0)         |           |
| Gliosis                                                                                                                          | 13  | 7 (53.8)         |           |
| Others                                                                                                                           | 26  | 12 (56.2)        |           |
| No specific finding                                                                                                              | 8   | 1 (12.5)         |           |
| EEG, electroencephalography; No., number of patients; MRI, magnetic resonance imaging.<br>*Statistically significant : p < 0.05. |     |                  |           |

**Table 2. Surgical outcome and the extent of resection**

| Extent of electrode resection | No. | Seizure-free (%) | p-Value |
|-------------------------------|-----|------------------|---------|
| All patients                  | 177 |                  |         |
| Group 1                       | 73  | 18 (24.7)        | <0.001* |
| Group 2                       | 21  | 12 (57.1)        |         |
| Group 3                       | 83  | 45 (54.5)        |         |
| Patients with TLE             | 66  |                  |         |
| Group 1                       | 22  | 8 (36.4)         | 0.778   |
| Group 2                       | 9   | 4 (44.4)         |         |
| Group 3                       | 35  | 16 (45.7)        |         |
| Patients with extra-TLE       | 111 |                  |         |
| Group 1                       | 51  | 10 (19.6)        | <0.001* |
| Group 2                       | 12  | 8 (66.7)         |         |
| Group 3                       | 48  | 29 (60.4)        |         |

No., number of patients; TLE, temporal lobe epilepsy.

Group 1, patients who underwent resection including not all electrodes showing ictal discharges during the first 3 sec after ictal onset; group 2, patients who underwent resection including all electrodes which showed ictal discharges during the first 3 sec; group 3: patients who underwent resection including all electrodes which showed ictal discharges during the first 5 sec.

\*Statistically significant:  $p < 0.05$ .

outcomes of group I and those of the other two groups but no difference in those of group II and group III: 24.7% in group I, 54.5% in group II, and 57.1% in group III ( $p < 0.001$ ; Table 2). Subgroup analysis according to

**Table 3. Surgical outcome and various predictors**

|                                                               | No. | Seizure-free (%) | p-Value |
|---------------------------------------------------------------|-----|------------------|---------|
| Including pathologic delta waves                              | 98  |                  |         |
| Total                                                         | 50  | 29 (58.0)        | <0.001* |
| Partial or no                                                 | 47  | 11 (22.9)        |         |
| Including frequent interictal spikes (above $\frac{1}{5}$ Hz) | 89  |                  |         |
| Total                                                         | 45  | 22 (48.9)        | 0.02*   |
| Partial or no                                                 | 44  | 11 (25.0)        |         |
| Including spikes of >1 Hz                                     | 35  |                  |         |
| Total                                                         | 17  | 10 (58.8)        | 0.064   |
| Partial or no                                                 | 18  | 5 (27.8)         |         |
| Including spikes of 1 to $\frac{1}{3}$ Hz                     | 29  |                  |         |
| Total                                                         | 10  | 3 (30.0)         | 0.593   |
| Partial or no                                                 | 19  | 4 (21.1)         |         |
| Including spikes of above $\frac{1}{3}$ to $\frac{1}{5}$ Hz   | 25  |                  |         |
| Total                                                         | 18  | 9 (50.0)         | 0.332   |
| Partial or No                                                 | 7   | 2 (28.6)         |         |
| Including gamma burst                                         | 20  |                  |         |
| Total                                                         | 14  | 8 (57.1)         | 0.329   |
| Partial or no                                                 | 6   | 2 (33.3)         |         |
| No., number of patients.                                      |     |                  |         |
| *Statistically significant: p < 0.05.                         |     |                  |         |

location indicated that there was no relationship between the extent of resection and surgical outcome in neocortical TLE.

The factors influencing the extent of resection were investigated. Among the variables mentioned, both the distribution of ictal onset and the propagation speed were significantly associated with the extent of resection ( $p < 0.001$  for both). All patients with wide ictal onset and about 80% of patients with rapid propagation belonged to group I and neither ictal-onset distribution nor propagation speed affected the surgical outcome in this group. Laterality, the location of ictal onset, the presence or absence of an MRI lesion, and pathology had no effect on the extent of resection.

Multivariate analysis indicated that both the extent of resection [ $p = 0.008$ , likelihood ratio (LR) 7.1] and the presence of a visible MRI lesion ( $p < 0.001$ , LR 12.7) were independent predictors of a seizure-free outcome.

### Surgical outcome and interictal abnormalities

We also analyzed the relationship between interictal abnormalities and surgical outcome (Table 3). First, in patients with pathologic delta waves ( $n = 98$ ), the total removal of the electrodes showing interictal pathologic delta waves predicted a seizure-free outcome (29 of 50 vs. 11 of 47,  $p < 0.001$ ). Moreover, patients who underwent the total removal of the pathologic delta waves had a better chance of a seizure-free outcome than did patients without pathologic delta waves ( $p = 0.007$ , not shown in the table). In patients with frequent interictal spikes ( $n = 89$ ), the total removal of electrodes showing frequent interictal spikes was significantly associated with a good surgical outcome

(22 of 45 vs. 11 of 44,  $p = 0.02$ ), and the removal of electrodes showing interictal spikes was closely influenced by the presence of ictal discharges ( $p < 0.001$ ). In patients with interictal spikes in concordance with ictal discharges ( $n = 36$ ), the electrodes showing interictal spikes were totally removed in all the patients. However, in the others, only 11 of 53 patients displayed the total removal of electrodes showing interictal spikes. We found no significant relationship between the removal of areas showing interictal spikes with different frequencies and the surgical prognosis. In patients with paroxysmal fast activity ( $n = 20$ ), the total removal of the electrodes with gamma bursts did not affect the surgical outcome.

## DISCUSSION

In this retrospective study, we found that the extent of resection was associated with the surgical outcome and that total resection, including all electrodes with ictal discharges during the first 3 sec, was a specific predictor of a good surgical outcome. Among the intracranial EEG findings, the total resection of electrodes displaying interictal pathologic delta waves or frequent interictal spikes was another strong predictor of a good surgical outcome ( $p = 0.001$  and  $p = 0.02$ , respectively). Neither the distribution of ictal onset nor the propagation speed influenced the surgical outcome in patients with narrow neocortical resection. The presence of a visible lesion on MRI was significantly associated with a good surgical outcome ( $p < 0.001$ ), which is consistent with the results of previous studies (Hennessy et al., 2001; Tassi et al., 2002; Cohen-Gadol et al., 2004; Tonini et al., 2004; Yun et al., 2006; Fauser et al., 2008).

A meta-analysis of the predictors of surgical outcome (Tonini et al., 2004) reported that various studies have confirmed a better surgical outcome after extensive resection than after limited resection. A recent study of subdural electrode analysis of cortical dysplasia (Widdess-Walsh et al., 2007) demonstrated that ictal onset at the edge of the subdural electrode coverage and incomplete resection of ictal epileptiform abnormalities predicted an increased risk of seizure recurrence. Another study also demonstrated that the complete resection of the ictal fields mapped with subdural electrode arrays gave a better prognosis than did partial resection (Jennum et al., 1993). We attempted to define more quantitatively the extent of resection that would achieve a good surgical outcome by dividing patients into three groups according to the extent of their resection. Our results should be helpful in deciding the extent of neocortical resection.

Subgroup analysis showed no relationship between the extent of resection and surgical outcome in neocortical TLE. In the meta-analysis discussed above, there was heterogeneity in the results assessing the extent of surgical resection and surgical outcome, and the authors suggested that the extent of resection may be affected by the underlying

pathology and the site of surgery (Tonini et al., 2004). Although we tried to include only neocortical TLE patients in the present study, it is possible that some patients with “undetected” hippocampal epileptogenicity could be included in the present study, considering the most common pathology in our study was focal cortical dysplasia and the temporal cortical dysplasia is frequently associated with hippocampal sclerosis (Srikijvilaikul et al., 2003; Fauser et al., 2004), and it can in part explain the lack of relationship between the extent of resection and surgical outcome in the neocortical TLE subgroup.

It is important to note that the temporal lobe is a confined structure compared with the other lobes and, unlike mTLE, extra-TLE does not have a consistent epileptogenic network or spreading pathway. The epileptogenic network of extra-TLE is unknown, but based on our results, the initial ictal spreading area and the area with interictal abnormalities on intracranial EEG may be components of the epileptogenic network. There are a number of studies on the epileptogenic network in mTLE using various methods (Henry et al., 1993; Bertram et al., 1998; Bartolomei et al., 2001; Bernasconi et al., 2004; Mueller et al., 2004; Lin et al., 2007; Bartolomei et al., 2008). In these studies, epileptogenic zone giving rise to initial ictal discharge was more accurately defined as a network of neuronal structures rather than a focus of abnormal activity. The high epileptogenicity of hippocampus is generally accepted in patients with mTLE, so previous studies have focused on the epileptogenicity of parahippocampal areas and their connectivity with hippocampus. On the contrary, the analysis of epileptogenic zone in neocortical epilepsy is more complicated because of the heterogeneity of involved area and the underlying pathology. Moreover, the determination of epileptogenic network in neocortical epilepsy generally depends on proper electrode placement, but intracranial EEG monitoring was performed as a part of presurgical investigation, so the selection of patients and the implantation of electrodes were performed strictly according to the clinical needs of exact determination of the ictal-onset zone and eloquent area but not for a complete analysis of the neocortical epileptogenic network. It would be an important finding if we could map the epileptogenic network in neocortical epilepsy, but we think more elaborate design and analysis of intracranial electrical activity would be necessary to define the epileptogenic zone in neocortical epilepsy.

Our intracranial EEG findings indicated that two ictal-onset morphologies—sinusoidal waves and spikes or sharp waves—did not influence the surgical outcome. Furthermore, there was no significant difference in frequency between the three categories, although ictal onset in the beta or gamma range was more strongly associated with a good surgical outcome. The presence of focal high-frequency oscillations near the time of seizure onset may signify a proximity to the epileptogenic focus (Jirsch et al., 2006). In contrast, another study reported that patients with slow

ictal-onset rhythms or repetitive sharp waves had a greater chance of being seizure-free than patients with low-voltage fast activity (Jung et al., 1999). Therefore, it remains unclear whether a specific ictal-onset morphology can predict a better surgical outcome.

As well as ictal intracranial EEG data, we investigated various interictal abnormalities. A resection that included frequent interictal spikes ( $> \frac{1}{3}$  Hz) predicted a good surgical outcome. In these results, a concordance with the ictal discharges was important in deciding whether the electrodes with interictal spikes should be removed. Several authors have postulated that interictal epileptiform discharges are associated with good surgical outcomes in neocortical or medial TLE (Chee et al., 1993; Bautista et al., 1999; Lee et al., 2005; Stefan et al., 2008). The study of Bautista et al. demonstrated that the presence of interictal epileptiform discharges extending beyond the area of resection correlated with poor surgical outcomes in patients with extrahippocampal epilepsy. Therefore, they suggested that the spatial extent of the interictal epileptiform discharges may be a better estimate of the extent of the epileptogenic zone and should be used to determine the limits of surgical resection (Bautista et al., 1999). However, many are reluctant to rely on interictal spikes alone in planning surgery (Sperling, 2003). For example, multifocal spikes are common, and even though a structural epileptogenic lesion may be present focally, interictal spikes can be widespread in one or both hemispheres. Therefore, interictal spikes are most useful when considered in conjunction with ictal EEG findings (Sperling, 2003).

We also analyzed the relationship between interictal pathologic delta waves and surgical outcome, and the total removal of electrodes with pathologic delta waves resulted in a good surgical outcome. Therefore, pathologic delta waves can be one of the markers of local epileptogenicity in neocortical epilepsy. Furthermore, continuous focal delta activity strongly indicates the presence of underlying pathologic lesions. Not only the removal of the ictal-onset zone, but the removal of any pathologic lesion is important for a good prognosis in epilepsy surgery. In another study, interictal slow wave activities in the delta and theta bands were more frequent in the Engel I outcome group than in Engel II–IV group (Stefan et al., 2008). Focal spike activities and focal slowing represent functional disturbances, and they can provide additional information that can be used to define the extent of the resection.

Interictal or ictal paroxysmal fast activity has been proposed as a surrogate marker of epileptogenic networks (Rampp & Stefan, 2006; Widdess-Walsh et al., 2007; Jacobs et al., 2008). In our study, patients who underwent complete resection of fast activity had a better chance of a seizure-free outcome (57.1%, 8 of 14) than patients who underwent incomplete resection (33.3%, 2 of 6). However, this result was not statistically significant, also because the number of patients was small. In a previous study with

focal cortical dysplasia (Widdess-Walsh et al., 2007), paroxysmal fast activity and slow runs of repetitive spikes were correlated closely with the ictal-onset zones. In our study, total removal of electrodes showing frequent interictal spikes was associated with a good surgical outcome, but we could not find any prognostic role of paroxysmal fast activity. We think this lack of association may be due to little overlap of interictal spikes and paroxysmal fast activity. Although there were overlap areas between these interictal patterns, repetitive interictal spikes were observed more frequently and the areas of interictal spikes were broader than those of paroxysmal fast activity in our study. To summarize, areas with both ictal onset and interictal abnormalities on intracranial EEG should be included in the limits of surgical resection if they are not located around the eloquent area.

Intracranial EEG monitoring was performed when other diagnostic modalities were nonconclusive or incongruent, and actually all patients with normal MRI who underwent surgical treatment underwent intracranial EEG monitoring. This particular indication of intracranial EEG monitoring in part explains the low diagnostic sensitivity of MRI in the present study and the apparent no outcome difference between the different pathologic subtypes. It is generally accepted that the surgical outcome of focal cortical dysplasia is no better than those of other pathologies, because the demarcation of the lesion is frequently poor and MRI frequently does not show any abnormalities in patients with pathologically proven focal cortical dysplasia (Widdess-Walsh et al., 2006). However, we believe that recruitment of a large number of patients may confirm the prognostic value of pathologic subtypes, even in patients who underwent intracranial EEG monitoring.

In conclusion, resection that includes more electrodes with ictal rhythm or interictal abnormalities predicts a good surgical outcome. However, it must be considered that the present study was retrospective design with a possibility of selection bias, and that more research is required to establish the surrogate intracranial EEG markers for the epileptogenic area to facilitate decisions regarding the extent of electrode resection.

## DISCLOSURE

The first two authors contributed equally to this work. We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines. None of the authors has any conflict of interest to disclose.

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